Initial Post

by Abdulla Husain Salem Hadna Almessabi - Thursday, 28 August 2025, 7:34 PM
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The ACLs, including KQML, are at the centre of letting autonomous agents share information and coordinate their actions. A key benefit of ACLs is the ability to offer a high-level, standardised communication protocol that is independent of the implementation language. This renders them appropriate in heterogeneous and distributed settings where agents written in the various platforms need to interact (Finin, McKay, and Fritzson, 1992). ACLs can be extended with richer interactions such as negotiation, cooperation, and delegating tasks (which are imperative to intelligent multi-agent systems) by supporting richer interactions, such as ask, tell, or achieve (Hai-long and Tie-jun, 2001). ACLs have a sound semantic basis as they can be traced to AI research, representing ideas in the knowledge representation, reasoning, and speech act theory (Chalupsky et al., 1992).

ACLs have flaws as well. They are usually expressive and abstract at the expense of efficiency. High-level communication acts can be computationally expensive to parse, interpret, and reason about in comparison to low-level mechanisms. Also, true semantic interoperability is still out of reach because other agents could read performatives differently unless an ontology is agreed upon (Hendler and Saltz, 1995). This is the opposite of the method invocation in languages as Python or Java, interaction is direct and tightly coupled, and more efficient. Method invocation can guarantee syntactic and semantic correctness in one runtime, but does not offer the character and openness of ACL-based communication between distributed systems (Luo, Zou, and Luo, 2012).

KQML and other ACLs have flexibility, interoperability, and AI-based communication of intelligent systems, whereas method invocation is simple, efficient, and understandable in controlled settings (Mayfield, Labrou, and Finin, 1995). Whether the problem involves a need to have heterogeneous multi-agent coordination or intra-language software integration is a key factor in the selection between them (Drasko, and Rakic, 2024).

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